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Assignment 7A

Aim
Write a menu driven program to implement stack using linked list.

Theory
Stack is a linear data structure which is implemented using array or linked list.

Push (Insert)
It adds an element to the start of the linked list (stack). The node of the element points to the element at the start ~~of~~ in the linked list.

POP (Delete)
It removes an element at the top of the stack which is the first element of the linked list and the node is freed.

PEEK (Top element)
It displays the top element of the stack that is the first element of the linked list

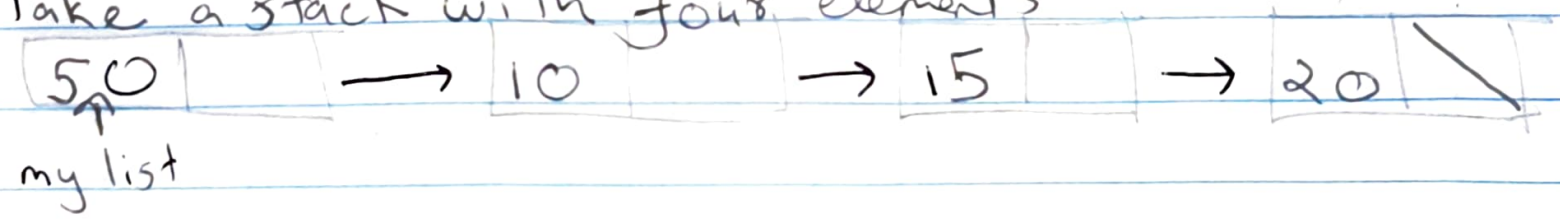
Time complexity
The time complexity is O(n) as all elements need to be traversed.

Conclusion

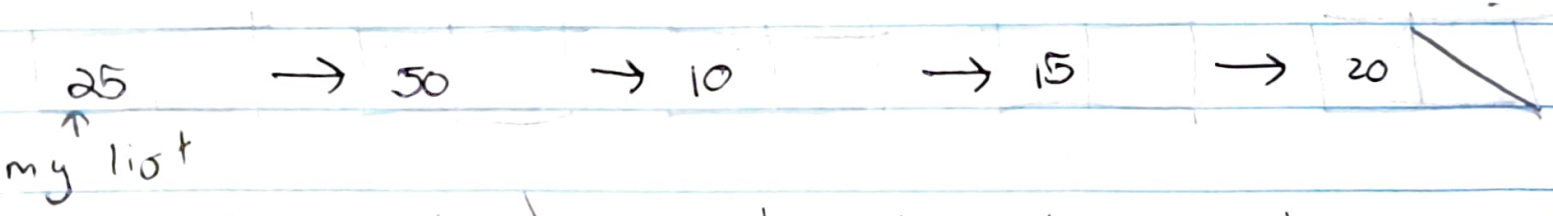
The concept of linked lists is applied to evaluate problems in a variety of applications.

Working Example

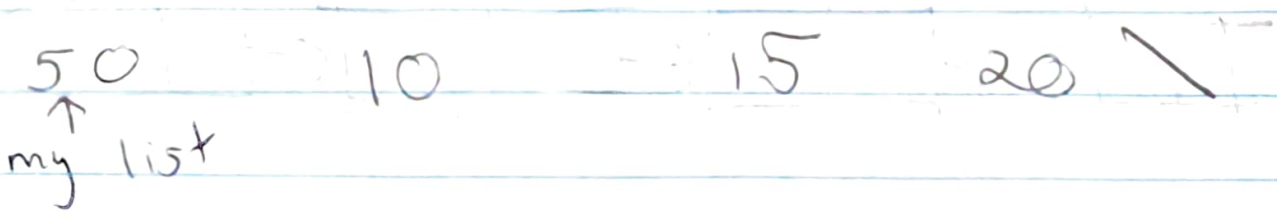
Take a stack with four elements



Now to push an element (25) insert it in the beginning and point the node to the first



Now to pop (25), point my list to 50 and free 25



```
#include<stdio.h>
#include<conio.h>
#include<malloc.h>
struct ll_node
{
   int data;
   struct ll_node *next;
};
struct ll_node *push(struct ll_node *top1,int val);
struct ll_node *pop(struct ll_node *top1);
void display(struct ll_node *top1);

void main() {
   int val, choice;
   struct ll_node *top=NULL;
   do
   {
      printf("----------------MENU----------------\n");
      printf("1.PUSH\n2.POP\n3.DISPLAY\n4.EXIT\nEnter your choice:");
      scanf("%d",&choice);
      switch(choice)
      {
         case 1:printf("Enter the value to be pushed:");
         scanf("%d",&val);
         top=push(top,val);
         break;
         case 2:
         top=pop(top);
```

```
        break;
        case 3:
        display(top);
        break;
        case 4:printf("......exiting the program\n");
        break;
        default: printf("INVALID CHOICE\n");
      }
   }while(choice!=4);
}

struct ll_node *push(struct ll_node *top1,int val) {
   struct ll_node *pnew=(struct ll_node *)malloc(sizeof(struct ll_node));
   pnew->next=NULL;
   pnew->data=val;
   pnew->next=top1;
   top1=pnew;
   return top1;
}

struct ll_node *pop(struct ll_node *top1) {
   struct ll_node *node2delete;
   if(top1==NULL)
   {
      printf("STACK UNDERFLOW\n");
      return top1;
   }
   node2delete=top1;
```

```
    top1=top1->next;
    free(node2delete);
    return top1;
}

void display(struct ll_node *top1) {
    struct ll_node *help_pointer=top1;
    if(top1==NULL) {
        printf("STACK UNDERFLOW\n");
        return;
    }
    printf("Stack contains:\n");
    while(help_pointer!=NULL) {
        printf("%d\t",help_pointer->data);
        help_pointer=help_pointer->next;
    }
    printf("\n");
    return;
}
```

```
---------------MENU----------------
1.PUSH
2.POP
3.DISPLAY
4.EXIT
Enter your choice:1
Enter the value to be pushed:10
---------------MENU----------------
1.PUSH
2.POP
3.DISPLAY
4.EXIT
Enter your choice:1
Enter the value to be pushed:20
---------------MENU----------------
1.PUSH
2.POP
3.DISPLAY
4.EXIT
Enter your choice:1
Enter the value to be pushed:30
---------------MENU----------------
1.PUSH
2.POP
3.DISPLAY
4.EXIT
Enter your choice:3
Stack contains:
30      20      10
```

```
---------------MENU----------------
1.PUSH
2.POP
3.DISPLAY
4.EXIT
Enter your choice:2
---------------MENU----------------
1.PUSH
2.POP
3.DISPLAY
4.EXIT
Enter your choice:3
Stack contains:
20      10
```